Weihao Liu

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EDUCATION

University of Michigan, USA Aug 2022 – 2026/2027 (Expected)
Ph.D. in Climate and Space Sciences and Engineering GPA: 4.0/4.0
Ph.D. in Scientific Computing

University of Science and Technology of China (USTC), China Aug 2018 – Jul 2022
B.Sc. in Geophysics GPA: 4.11/4.3
B.Eng. in Computer Science and Technology

SELECTED AWARDS

- **The 2025–2026 MICDE Fellowship**, Michigan Institute for Computational Discovery and Engineering, University of Michigan, August 2025.
- **The 2022–2023 Franklin D. and Clarence T. Johnston Fellowship**, College of Engineering, University of Michigan, September 2022.
- **The 2021–2022 Guo Moruo Scholarship (Highest Honor for undergraduates at USTC)**, USTC, April 2022.
- **The 2019–2020, 2020–2021 National Scholarship (Top 1%)**, USTC, October 2019, 2020.

RESEARCH INTERESTS

- Solar and heliospheric physics, solar energetic particles, galactic cosmic rays, and particle radiation transport.
- Computational physics, scientific computing, and machine learning for space-weather applications.

PUBLICATIONS

First-Author Papers

- **Weihao Liu**, Xianyu Liu, David Lario, Lulu Zhao, Tamas I. Gombosi, Alexander D. Shane, and Igor V. Sokolov. *arXiv* **2606.02445**; *ApJ* (Submitted, 2026). **Link**
- **Weihao Liu**, Lulu Zhao, Igor V. Sokolov, Kathryn Whitman, Tamas I. Gombosi, Nishtha Sachdeva, Eric T. Adamson, Hazel M. Bain, Claudio Corti, M. Leila Mays, Michelangelo Romano, Carina R. Alden, Madeleine M. Anastopulos, Mary E. Aronne, Janet E. Barzilla, Wesley T. Cook, Shawn D. Dahl, Hannah Hermann, Anthony J. Iampietro, A. Steve Johnson, Elizabeth A. Juelfs, Melissa R. Kane, Jonathan D. Lash, Kimberly Moreland, Briana K. Muhlestein, Teresa Nieves-Chinchilla, Edward Semones, James F. Spann, Earl M. Spencer, Luke A. Stegeman, Christopher J. Stubenrauch, and Kenneth L. Tegnell. *Space Weather* **24**, 3 (2026). **Link**
- **Weihao Liu**, Mikhail Dobynde, Jingnan Guo, Jordanka Semkova, and Krasimir Krastev. *arXiv* **2511.09040**; *Life in Space Sciences and Research* (In Revision, 2026). **Link**
- **Weihao Liu**, Igor V. Sokolov, Lulu Zhao, Tamas I. Gombosi, Nishtha Sachdeva, Xiaohang Chen, Gábor Tóth, David Lario, Ward B. Manchester IV, Kathryn Whitman, Christina M. S. Cohen, Alessandro Bruno, M. Leila Mays, and Hazel M. Bain. *ApJ* **985**, 82 (2025). **Link**
- **Weihao Liu**, Jingnan Guo, Yubao Wang, and Tony C. Slaba. *ApJS* **271**, 18 (2024). **Link**
- **Weihao Liu**, Jingnan Guo, Jian Zhang, and Jordanka Semkova. *ApJ* **949**, 77 (2023). **Link**

Co-Author Papers

- Juan G. Alonso Guzmán, **Weihao Liu**, Vladimir Florinski, Gábor Tóth, Keyvan Ghanbari, and Bart van der Holst. *ApJ* (Submitted, 2026).
- Peijin Zhang, **Weihao Liu**, Bin Chen, Ward B. Manchester IV, Surajit Mondal, and Sijie Yu. *ApJ* (Submitted, 2026).
- Chao Zhang, **Weihao Liu**, Jingnan Guo, Mikhail Dobynde, and Robert F. Wimmer-Schweingruber. *Space Weather* **24**, 3 (2026). **Link**

- David Lario, Junxiang Hu, Laura Balmaceda, Ryun-Young Kwon, Samantha Wallace, Daniel Da Silva, M. Leila Mays, Vratislav Krupar, Fernando Carcaboso, Alexander Warmuth, Muriel Stiefel, Leng Ying Khoo, Ian Richardson, Patrick Kühl, Robert F. Wimmer-Schweingruber, Rami Vainio, Esa Riihonen, **Weihao Liu**, Angels Aran, Evangelos Paouris, Melissa Kane, Christina O. Lee, Clementina Sasso, Charles Arge, Shaela I. Jones, Jaye Verniero, Antonio E. Niemela, Timothy Horbury, and Milan Maksimovic. *ApJ* **998**, 260 (2026). **Link**
- Xianyu Liu, **Weihao Liu**, Ward B. Manchester IV, Daniel T. Welling, Gábor Tóth, Tamas I. Gombosi, Marc L. DeRosa, Luca Bertello, Alexei A. Pevtsov, Alexander A. Pevtsov, Kevin Reardon, Kathryn Wilbanks, Amy Rewoldt, and Lulu Zhao. *ApJ* **997**, 243 (2026). **Link**
- Martin A. Reiss, Maria M. Kuznetsova, Claudio Corti, Jia Yue, et al. (including **Weihao Liu**). *ESSOAr*; *Space Weather* (Accepted, 2026). **Link**
- Claudio Corti, Maria M. Kuznetsova, Martin A. Reiss, Jia Yue, et al. (including **Weihao Liu**). *ESSOAr*; *Space Weather* (Accepted, 2026). **Link**
- Ke Hu, Kevin Jin, Victor Verma, **Weihao Liu**, Ward B. Manchester IV, Lulu Zhao, Tamas I. Gombosi, and Yang Chen. *arXiv* **2512.13417**; *Space Weather* (Submitted, 2026). **Link**
- Xiaohang Chen, Lulu Zhao, Joe Giacalone, Nishtha Sachdeva, Igor V. Sokolov, Gábor Tóth, Christina M. S. Cohen, David Lario, Fan Guo, Athanasios Kouloumvakos, Tamas I. Gombosi, Zhenguang Huang, Ward B. Manchester IV, Bart van der Holst, **Weihao Liu**, David J. McComas, Matthew E. Hill, and George C. Ho. *ApJ* **994**, 242 (2025). **Link**
- Yubao Wang, Jingnan Guo, Claudio Corti, Yuming Wang, **Weihao Liu**, and Robert F. Wimmer-Schweingruber. *A&A* **704**, A314 (2025). **Link**
- Lulu Zhao, Igor V. Sokolov, Tamas I. Gombosi, David Lario, Kathryn Whitman, Zhenguang Huang, Gábor Tóth, Ward B. Manchester IV, Bart van der Holst, Nishtha Sachdeva, and **Weihao Liu**. *Space Weather* **22**, 9 (2024). **Link**
- Jingnan Guo, Cary Zeitlin, Robert F. Wimmer-Schweingruber, Donald M. Hassler, Bent Ehresmann, Scot Rafkin, Johan L. Freiherr von Forstner, Salman Khaksarighiri, **Weihao Liu**, and Yuming Wang. *A&A Review*, **29**, 1–81 (2021). **Link**

PRESENTATIONS

Selected Oral Presentations

- Simulated Real-Time Testing of the Prototype Implementation of the SOFIE Model: The 2025 Space Weather Prediction Testbed Exercise, EGU 2026, Vienna, Austria, May 3–8, 2026. **Link**
- Operational Testing of Physics-Based SEP Prediction Model, SOFIE, in the SWPC Testbed Exercise, AGU 2025, New Orleans, LA, USA, December 15–19, 2025. **Link**
- Simulated Real-Time Testing of the Prototype Implementation of the SOFIE Model: The 2025 Space Weather Prediction Testbed Exercise, DASH 2025, San Antonio, TX, USA, October 19–24, 2025. **Link**
- Simultaneous Simulation of the Solar Energetic Proton Acceleration and the Forbush Decrease in Galactic Cosmic Ray Fluxes Following the July 14, 2017 Event, 39th ICRC 2025, Geneva, Switzerland, July 14–24, 2025. **Link**
- SOFIE and Its Application to Modeling the 2017 September 10 Solar Energetic Particle Event as Observed on Mars and at 1 au, ASTRONUM 2025, Madison, WI, USA, July 14–18, 2025. **Link**
- Not All Shocks Are Created Equal: Shock Acceleration During the 2013 April 11 Solar Energetic Particle Event, EGU 2025, Vienna, Austria, April 27–May 2, 2025. **Link**
- Physics-Based Simulation of the 2013 April 11 Solar Energetic Particle Event, ISWAT 2025, Port Canavera, FL, USA, February 10–14, 2025.
- Solar Energetic Particle Benchmark Dataset in the CLEAR Space Weather Center of Excellence, SEP Monitoring and Forecasting Workshop, Atlanta, GA, USA, October 16–19, 2025.
- Physics-Based Simulation of the 2013 April 11 Solar Energetic Particle Event, SEP Monitoring and Forecasting Workshop, Atlanta, GA, USA, October 16–19, 2025.
- Multi-Spacecraft Modeling of Solar Energetic Particle Events Using Solar Wind with Field Lines and Energetic Particles, AGU 2023, San Francisco, CA, USA, December 11–15, 2023. **Link**

Selected Poster Presentations

- Counterintuitive Magnetic Connectivity and Energetic Particle Flux Differences among Nearby Observers During the 2023 February 24 Solar Energetic Particle Event, SHINE 2026, Madison, WI, USA, June 28–July 2, 2026. **Link**
- Modeling the 2017 September 10 Solar Energetic Particle Event as Observed on Mars and at 1 au, AGU 2025, New Orleans, LA, USA, December 15–19, 2025. **Link**
- Modeling the 2017 September 10 Solar Energetic Particle Event as Observed on Mars and at 1 au, SHINE 2025, Charleston, SC, USA, June 23–27, 2025. **Link**
- Solar Energetic Particle Benchmark Dataset in the CLEAR Space Weather Center of Excellence, AGU 2024, Washington, D.C., USA, December 9–13, 2024. **Link**
- High-Resolution Poisson Bracket Scheme Applied to Simulate the 2013 April 11 Solar Energetic Particle Event, AGU 2024, Washington, D.C., USA, December 9–13, 2024. **Link**
- Simulating Solar Energetic Particle Events Using a Higher-Order TVD Poisson Bracket Scheme in SWMF, SHINE 2024, Juneau, AK, USA, August 12–16, 2024. **Link**
- Benchmarking Dataset of Solar Energetic Particle Events: Validation, Analysis and Implications, AGU 2023, San Francisco, CA, USA, December 11–15, 2023. **Link**
- A Comprehensive Comparison of Various Galactic Cosmic Ray Models, AGU 2023, San Francisco, CA, USA, December 11–15, 2023. **Link**
- A Benchmark Solar Energetic Particle Events Dataset, SHINE 2023, Stowe, VT, USA, August 7–11, 2023. **Link**

TECHNICAL SKILLS

Programming Languages	Fortran, Python, C, C++, Julia, Perl, Bash
Scientific Computing	OpenMP, MPI, SLURM, PETSc, Linux/Unix, NumPy, SciPy, Pandas
Technical Software	Git, IDL, MATLAB, Tecplot, LaTeX, MySQL, Jupyter, OpenAI CodeX

MENTORING EXPERIENCE

- Juno Larson (PhD Student; Year 2), *Modeling Extreme SEP Events within the SOFIE Model* — May 2026 – Now
- Keheng Zhu (PhD Student; Year 1), *A One-Click SWMF–AI Agent for Physics-Based Simulations* — Apr 2026 – Now
- Gergely Koban (PhD Student; Year 1), *Automating the MHD–SEP Simulation Pipeline of SOFIE* — Oct 2025 – Now
- Nikolett Biró (PhD Student; Years 1–2), *A Magnetic Connectivity Tool with M-FLAMPA* — Mar 2025 – Now
- Chao Zhang (Undergraduate), *Radiation Dose Calculations for Earth–Mars Travel* — Feb 2025 – Mar 2026
- Ke Hu (Master's Student), *Assessing Solar Flare Data Sources for Machine Learning Forecasting* — Feb 2024 – May 2025
- Chia-Yun Li (Master's Student), *Machine Learning Forecasts of SEP Events at Earth* — Feb 2024 – May 2025
- Keheng Zhu (Undergraduate), *Bridging Chromosphere and Solar Corona Modeling in SWMF* — Jun – Oct 2024
- Alexander Trintchouk (High School Student), *Analysis and Calibration of SEP Observations* — May – Aug 2024

TEACHING EXPERIENCE

- Teaching Assistant in *Mathematical Analysis (B2)*, USTC — Feb – Jul 2022
- Teaching Assistant in *Fluid Mechanics*, USTC — Sep 2021 – Jan 2022
- Teaching Assistant in *Equations of Mathematical Physics (B)*, USTC (Excellent TA) — Feb – Jun 2021
- Teaching Assistant in *Functions of a Complex Variable (B)*, USTC — Sep – Dec 2020

PROFESSIONAL MEMBERSHIPS

- Student Member of the *American Astronomical Society* (AAS) — Since 2025
- Student Member of the *European Geosciences Union* (EGU) — Since 2025
- Student Member of the *American Geophysical Union* (AGU) — Since 2023